

ELEC/COEN 390

Engineering team design project

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Outline

- ▶ Introduction to SQL
- ▶ Using SQLite with Android

What is SQL?

- ▶ SQL stands for Structured Query Language
- ▶ SQL lets you access and manipulate databases
- ▶ SQL is an ANSI (American National Standards Institute) standard

What Can SQL do?

- ▶ SQL can execute queries against a database
- ▶ SQL can retrieve data from a database
- ▶ SQL can insert records in a database
- ▶ SQL can update records in a database
- ▶ SQL can delete records from a database
- ▶ SQL can create new databases
- ▶ SQL can create new tables in a database
- ▶ SQL can create stored procedures in a database
- ▶ SQL can create views in a database
- ▶ SQL can set permissions on tables, procedures, and views

How to think about a DB?

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- ▶ A DB is made of tables and each table has columns & records.
(A record is a row and every col represents a given data type)

Key	Name	Id	grade
1	Tawfiq	666	A+
2	Sara	123	C
3	Dr. Kharma	1100	F

Some of The Most Important SQL Commands

- ▶ **SELECT** - extracts data from a database
- ▶ **UPDATE** - updates data in a database
- ▶ **DELETE** - deletes data from a database
- ▶ **INSERT INTO** - inserts new data into a database
- ▶ **CREATE DATABASE** - creates a new database
- ▶ **ALTER DATABASE** - modifies a database
- ▶ **CREATE TABLE** - creates a new table
- ▶ **ALTER TABLE** - modifies a table
- ▶ **DROP TABLE** - deletes a table

SQLite

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- ▶ It is a software library that implements a **self-contained, serverless, zero-config, transactional SQL** DB engine.
- ▶ SQL (Structured Query Language)
- ▶ Comes packaged with Android as a C++ library.

SQLite

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- ▶ **self-contained:** resides within an application boundary.
- ▶ **serverless:** It does not require a separate server process or system to operate.
- ▶ **zero-config:** No setup or administration needed.
- ▶ **transactional:** A DBMS where write transactions on the db are able to be rolled back if they are not completed properly.
A transaction symbolizes a unit of work performed in a DBMS.
- ▶ A complete SQLite DB is stored in a single **cross-platform** (Unix and Windows) disk file.
- ▶ Simple, lightweight and easy-to-use API.

SQL Syntax

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- ▶ *Case insensitive* except for few commands.
- ▶ You can add *comments*, they start with "--"
i.e. >.help -- comment
- ▶ All SQL statements start with *keywords*.
- ▶ The ANSI standard prescribes a semicolon ";" to terminate a statement, although some systems support omitting it. Best practice is to always include the semicolon

Let's build 'PokeDex' DB with 2 tables

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PokeDex:	Pokemon	Moves
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As an example of what the 'Pokemon' table can be look at:

Key	Name	Type1	Type2	Height	Weight
1	Bulbasaur	GRASS	POISON	2.05	15.2
2	Ivysaur	GRASS	POISON	3.03	28.7
3	Venusaur	GRASS	POISON	6.07	220.5
4	Charmander	FIRE		2.00	18.7



Reference: <http://serebii.net/>

SQL CREATE/DROP DB

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- ▶ CREATE DATABASE <db_name>;
- ▶ DROP DATABASE <db_name>;

- ▶ ex:
- ▶ CREATE DATABASE PokeDex;
- ▶ DROP DATABASE PokeDex;

SQL CREATE TABLE

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- ▶ Create a table within your database
- ▶ `CREATE TABLE <table_name> (
 <field_name> <TYPE>, <field_name> <TYPE>);`
- ▶ ex:
`CREATE TABLE pokemon (
Name TEXT NOT NULL,
Type1 TEXT,
Type2 TEXT,
Height REAL,
Weight REAL
);`
- ▶ Type can be: TEXT, REAL, INTEGER, NULL, BLOB

SQL INSERT rows

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- ▶ Inserting an entry into the table, two ways
- ▶ Without specifying the column names (order matters)
 - ▶ INSERT INTO <table_name>
VALUES (<value>, <value>, ...);
 - ▶ ex:
INSERT INTO pokemon
VALUES ("Scizor","Bug","Steel",1.8,118);
- ▶ With specifying the column names (order doesn't matter)
 - ▶ INSERT INTO <table_name> (<col1_name>, <col2_name>,)
VALUES (<col1_value>, <col2_value>, ...);
 - ▶ ex:
INSERT INTO pokemon (weight, height, name,type2,type1)
VALUES (118, 1.8, "scizor", "Bug","Steel");

SQL SELECT rows

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- ▶ Now that you have data! You can select it

- ▶ Getting it all

```
SELECT (<col1_name>, <col2_name>, ....)  
FROM <table_name>;
```

- ▶ ex:

```
SELECT(Name, Type2, Type2)  
FROM Pokemon;
```

- ▶ With certain criteria

```
SELECT (<col1_name>, <col2_name>, ....)  
FROM <table_name>  
WHERE <col1_name> = <value> AND/OR <col2_name> = <value> <etc.>;
```

- ▶ ex:

```
SELECT(Name, Type2, Type2)  
FROM Pokemon  
WHERE Type1 = "Fire" AND Type2 = "Water";
```

- ▶ ORDER BY key DESC;

SQL UPDATE rows

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- ▶ This allows us to update existing rows in the database.

EXAMPLE

```
UPDATE <table_name>
```

```
SET <col1_name> = <value>, <col2_name> = <value> < etc.>
```

```
WHERE <col1_name> = <value> AND/OR <col2_name> = <value> < etc.>;
```

- ▶ ex:

```
UPDATE Pokemon
```

```
SET Type2 = "Water"
```

```
WHERE Type1 = "Fire";
```

SQL DELETE rows

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- ▶ This allows us to delete existing rows in the database.

EXAMPLE

```
DELETE FROM <table_name>  
WHERE <col1_name> = <value> AND/OR <col2_name> = <value> etc.;
```

- ▶ ex:
DELETE FROM Pokemon
WHERE Type1 != "Fire";

SQL ALTER a table

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- ▶ This allows us to alter existing tables in the database.
Add, drop, modify columns,

EXAMPLE

```
ALTER TABLE <table_name>  
ADD COLUMN <new_col_name> <new_col_type>;
```

- ▶ ex:
ALTER Pokemon
ADD COLUMN Area TEXT NOT NULL DEFAULT "Johto";

SQL DROP (Delete) a table

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- ▶ This allows us to delete tables in the database.

EXAMPLE

```
DROP TABLE <table_name>;
```

ex:

```
DROP TABLE Moves;
```

Demo: How to use SQLite with Android

Understanding Fragments

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- ▶ <https://developer.android.com/guide/components/fragments>
- ▶ DialogFragment reference:
- ▶ <https://developer.android.com/reference/android/support/v4/app/DialogFragment>

Questions?

References

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- ▶ http://www.w3schools.com/sql/sql_syntax.asp